

Understanding

1 Write the next three terms in these patterns.

- a $2^1, 2^2, 2^3, 2^4, \dots$
 b $x^2, x^3, x^4, \dots$
 c $3x, 3x^2, 3x^3, \dots$

2 Recall that $\frac{1}{4} = \frac{1}{2^2}$. Similarly write these using positive indices.

- a $\frac{1}{9}$ b $\frac{1}{25}$ c $\frac{5}{16}$ d $\frac{2}{27}$

3 Write each rule for negative indices by completing each statement.

- a $a^{-b} = \frac{1}{a^b}$ b $\frac{1}{a^b} = \frac{1}{a^b}$

Fluency

8 Express the following in simplest form with positive indices.

- a $x^2y^3 \times x^{-3}y^{-4}$ b $4a^6y^4 \times a^3y^2$ c $2a^3b \times 3a^2b^3$ d $6a^4b^3 \times 3a^{-6}b$
 e $a^3b^4 + (a^2b)^7$ f $p^2q^3 + (p^7q^2)$ g $\frac{a^4b^3}{a^2b^5}$ h $\frac{m^3n^2}{nm^3}$
 i $\frac{p^2q^4r^4}{pq^4r^5}$ j $\frac{3x^2y}{6xy^2}$ k $\frac{4m^3n^4}{7m^2n^7}$ l $\frac{12r^4s^6}{9rs^{-1}}$
 m $\frac{f^3g^2}{f^2g^3}$ n $\frac{r^3s^{-4}}{r^3s^{-2}}$ o $\frac{3w^2x^3}{6w^3x^{-2}}$ p $\frac{15c^2d}{12c^2d^{-3}}$

Example 18 9 Simplify the following and express your answers with positive indices.

- a $(a^3b^2)^3 \times (a^2b^4)^{-1}$ b $(2p^2)^4 \times (3p^2q)^{-2}$ c $2(x^2y^{-1})^2 \times (3xy^4)^3$
 d $\frac{2a^2b^2}{a^{-3}} \times \frac{2a^2b^5}{b^4}$ e $\frac{(3rs^2)^4}{r^3s^4} \times \frac{xy^4}{2x^2y}$ f $\frac{4(x^2y^4)^2}{x^2y^{-3}} \times \frac{xy^4}{2x^2y}$
 g $\left(\frac{a^2b^3}{b^2}\right)^2 + \left(\frac{ab^4}{a^2}\right)^{-2}$ h $\left(\frac{m^4n^2}{r^3}\right)^2 + \left(\frac{m^3n^2}{r^3}\right)^2$ i $\frac{3(x^2y^{-4})^2}{2(xy^2)^2} + \frac{(xy)^{-3}}{(3x^2y^4)^2}$

10 Evaluate without the use of a calculator.

- a 5^{-2} b 4^{-3} c 2×7^{-2} d $5 \times (-3)^{-4}$
 e $3^{10} \times (3^2)^6$ f $(4^2)^5 \times 4(4^{-3})^{-3}$ g $\frac{2}{7^{-2}}$ h $\frac{-3}{4^{-2}}$
 i $\left(\frac{2}{3}\right)^{-2}$ j $\left(\frac{-5}{4}\right)^{-3}$ k $\frac{(4^{-2})^3}{4^{-4}}$ l $\frac{(10^{-4})^2}{(10^{-2})^3}$

11 The width of a hair on a spider is approximately 3^{-5} cm. How many centimetres is this correct to four decimal places?



Fluency

Example 17a, b 4 Express the following using positive indices.

- a x^{-5} b a^{-4} c $2m^{-4}$ d $3y^{-7}$
 e $3a^2b^{-3}$ f $4m^{-3}n^{-3}$ g $10x^{-2}y^2z^5$ h $3x^{-4}y^{-2}z^3$
 i $\frac{1}{3}p^{-2}qr$ j $\frac{1}{5}d^{-2}e^{-4}f^5$ k $\frac{3}{8}a^{-2}b^{-6}c^{-7}$ l $\frac{2}{5}k^{-3}d^{-5}e^{-2}$

Example 17c 5 Express the following using positive indices.

- a $\frac{1}{x^2}$ b $\frac{y^3}{x}$ c $\frac{4}{m^{-7}}$ d $\frac{3}{b^3}$
 e $\frac{2b^4}{d^{-3}}$ f $\frac{3m^2}{n^{-4}}$ g $\frac{4b^4}{3a^{-3}}$ h $\frac{5h^3}{2g^{-3}}$

6 Use index laws 1 and 2 to simplify the following. Write your answers using positive indices.

- a $x^3 \times x^{-2}$ b $a^7 \times a^{-4}$ c $2b^5 \times b^{-9}$ d $3y^6 \times y^3$
 e $3a^{-4} \times 2a^2$ f $4x^{-5} \times 3x^4$ g $5m^{-4} \times (-2m^{-2})$ h $-3a^{-7} \times 6a^{-3}$
 i $\frac{2x^{-2}}{3x^{-3}}$ j $\frac{7d^{-3}}{10d^{-5}}$ k $\frac{6c^5}{12c^5}$ l $\frac{3b^2}{4b^{-4}}$
 m $\frac{5s^2}{3s}$ n $\frac{4f^{-5}}{3f^{-3}}$ o $\frac{3d^3}{6d^{-1}}$ p $\frac{15r^4}{18r^{-2}}$

Example 17d 7 Express the following with positive indices.

- a $\left(\frac{2x^2}{x^3}\right)^4$ b $\left(\frac{m^3}{4m^5}\right)^3$ c $2(x^{-7})^3$ d $4(d^2)^3$
 e $(3t^4)^2$ f $5(x^2)^2$ g $(3x^5)^4$ h $-8(x^5)^3$
 i $(4y^2)^{-2}$ j $(3h^3)^{-4}$ k $7(f^{-2})^4$ l $2(t^3)^{-2}$

12 a Simplify these numbers.

i $\left(\frac{2}{3}\right)^{-1}$ ii $\left(\frac{5}{7}\right)^{-1}$ iii $\left(\frac{2x}{y}\right)^{-1}$

b What is $\left(\frac{a}{b}\right)^{-1}$ expressed in simplest form?

13 A student simplifies $2x^{-2}$ and writes $2x^2 = \frac{1}{2x^2}$. Explain the error made.

14 Evaluate the following by combining fractions.

a $2^{-1} + 3^{-1}$ b $3^{-2} + 6^{-1}$ c $\left(\frac{3}{4}\right)^{-1} - \left(\frac{1}{2}\right)^0$
 d $\left(\frac{2}{3}\right)^{-1} - 5(2^2)$ e $\left(\frac{4}{5}\right)^{-2} - \left(\frac{2^2}{3}\right)$ f $\left(\frac{3}{2^2}\right)^{-1} - \left(\frac{2^1}{3^2}\right)^{-1}$

15 Prove that $\left(\frac{1}{2}\right)^x = 2^x$ giving reasons.

Enrichment: Simple equations with negative indices

16 Find the value of x.

a $2^x = \frac{1}{4}$ b $2^x = \frac{1}{32}$ c $3^x = \frac{1}{27}$ d $\left(\frac{3}{4}\right)^x = \frac{4}{3}$
 e $\left(\frac{2}{3}\right)^x = \frac{9}{4}$ f $\left(\frac{2}{5}\right)^x = \frac{125}{8}$ g $\frac{1}{2^x} = 8$ h $\frac{1}{3^x} = 81$
 i $\frac{1}{2^x} = 1$ j $5^{x-2} = \frac{1}{25}$ k $3^{x-3} = \frac{1}{9}$ l $10^{x-5} = \frac{1}{1000}$
 m $\left(\frac{3}{4}\right)^{2x+1} = \frac{64}{27}$ n $\left(\frac{2}{5}\right)^{3x-5} = \frac{25}{4}$ o $\left(\frac{3}{2}\right)^{3x+2} = \frac{16}{81}$ p $\left(\frac{7}{4}\right)^{1-x} = \frac{4}{7}$

Reasoning

Exercise 3H

1 a 2^3 b 16 c 2 d 3
 2 a 2^3 b 3^2 c 4^3 d 3^3
 3 a 2^3 b 3^2 c 4^3 d 3^3
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Exercise 3G

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